A TALK IN FOUR PARTS

The Anatomy of Interoperable Apps

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COLD OPEN

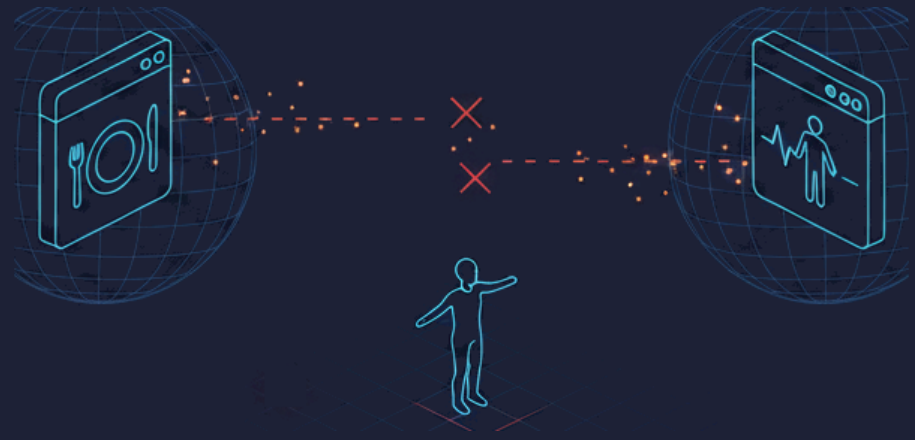
A simple question my apps cannot answer.

My partner has been dealing with pain that seems related to food.

The obvious question: **what did she eat, and when did the pain happen?**

Today, even if I build two perfect apps, they still do not naturally work together.

> Best food tracker on earth, plus best pain tracker on earth — and they still cannot answer this together.



THE DIAGNOSIS

Every app starts cold.

Every app you install today creates its own:

- Account
- Database
- Permission model
- Context

Every app acts like it is the center of the user's life.

That is why every new app starts cold.

A CLUE

Local agents already do this. They just cannot scale.

On a local machine, an agent can:

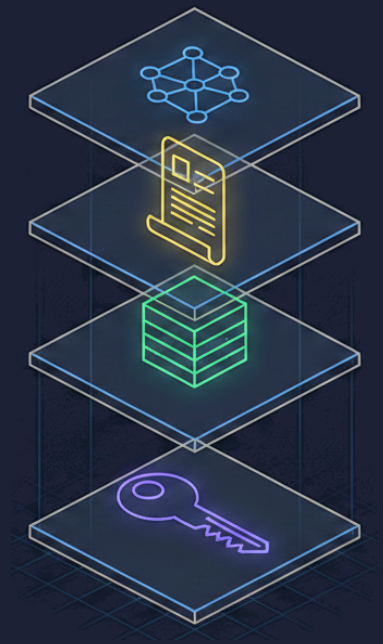
- Know it is working for you
- Read and write files in the same place
- Build apps that use data from another
- Create personalized tools on the fly

> This works because everything is local and implicitly trusted. **Powerful — but it does not scale** to real apps, real users, real permissions.

THE FRAME

What would it take to make that shape work on the internet?

Four parts. The rest of this talk goes through each one.



04 **Discovery**
How new apps find what is already there

03 **A contract**
What the app is allowed to do

02 **A place**
Where the data lives

01 **Identity**
Who is using the app

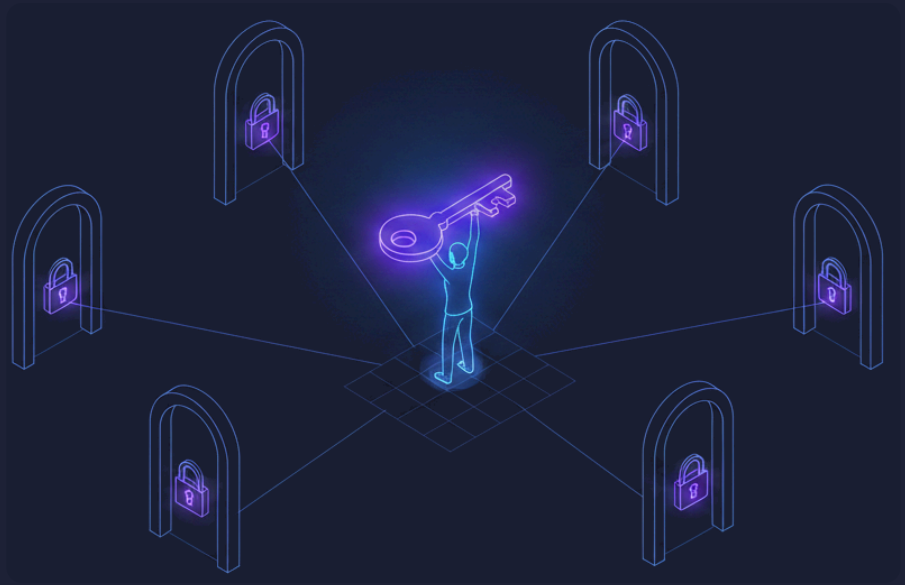
PART 1 OF 4 · IDENTITY

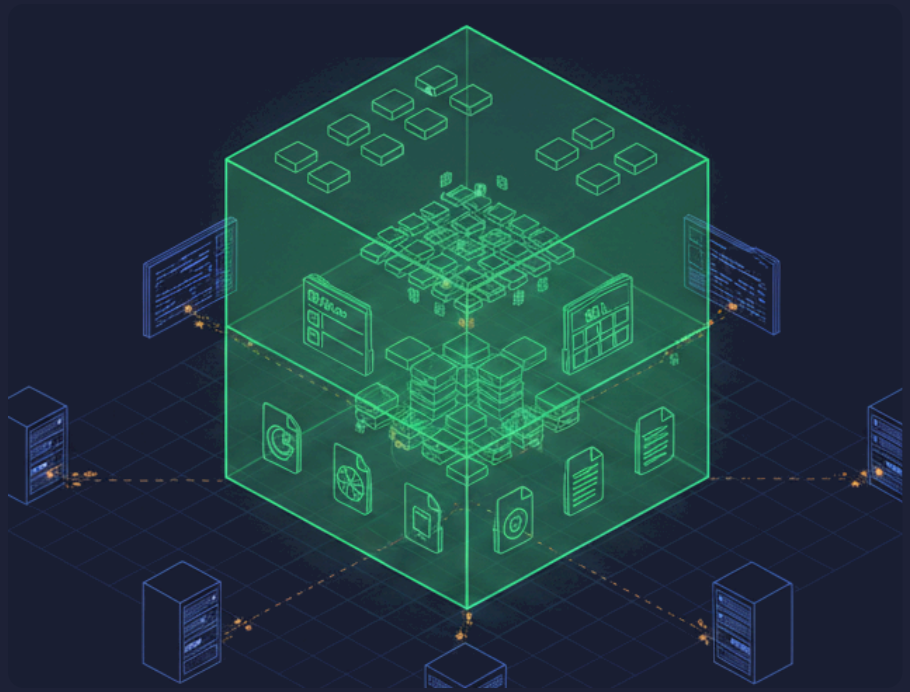
One person, many apps.
Not one account per app.

The app needs to know who is using it without forcing the user into yet another account.

- Passkey-based
- Biometric unlock
- Cryptographic identity
- No blockchain, no wallet

WebAuthn ships in every modern browser. The primitive already exists.





PART 2 OF 4 · A PLACE

Interoperability starts with a shared place to put data.

Shared identity is not enough. The apps need somewhere shared to put data.

- A user-owned space
- Key-value store + SQL store
- Apps write into the user's cloud, not their own silo

> Interoperability is not magic API glue. It starts with data living somewhere apps can jointly use it.

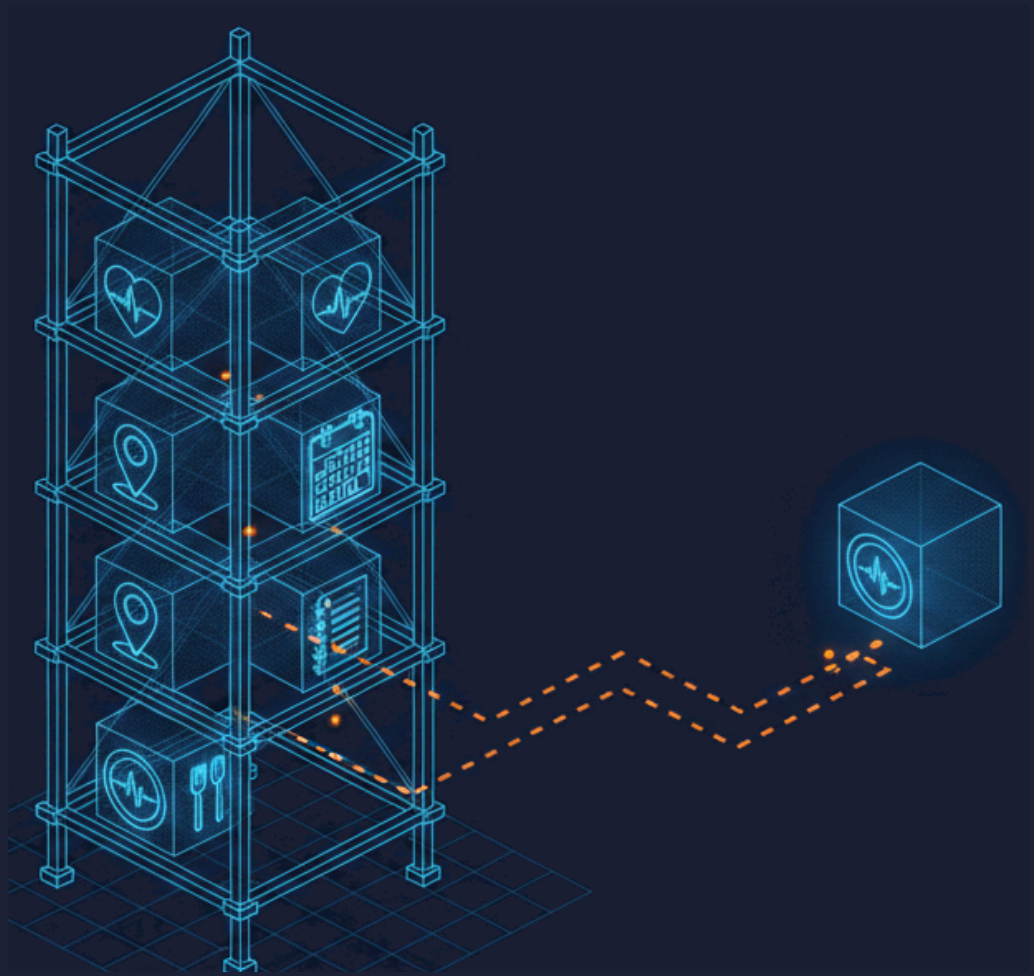
PART 3 OF 4 · A CONTRACT

The app does not ask for the kitchen sink. It declares what it needs.

If apps share a place, permissions become much more important.

A MANIFEST DECLARES

- **Reads** · what data the app touches
- **Writes** · what data the app produces
- **Delegations** · what it passes to other apps or agents
- **Approvals** · what the user must consent to



PART 4 OF 4 · DISCOVERY

Every new app enters a world that already has memory.

Once apps share a place, the next problem is: how does a new app or agent know what already exists?

A REGISTRY WHERE

- Apps publish their manifests
- Agents inspect what apps and data are available
- New apps inherit context instead of starting cold

Discovery is what makes the system compound.

DEMO

The four parts, working together.

Two apps, one user, no shared backend.

THE PUNCHLINE

These apps do not know about each other. They know about the user's data. That is enough.

The food tracker never called the pain tracker.

The pain tracker never called the food tracker.

The agent never called either of them.

The shared substrate did the work.

Composability without integrations. No partnerships. No API deals. No platform.

THE BIGGER BET

If users own the data, apps compete on quality instead of captivity.

App code is getting cheaper. Agents can clone or generate useful software quickly.

COMMODITIZING	APPRECIATING
The app shell	The user's context

The durable asset is not the app shell. It is the user's context.

WHERE WE COME IN

That is what we are building at TinyCloud.

Identity, storage, permissions, and discovery for interoperable apps.

IDENTITY	Passkey-based, biometric, no wallet	**OpenKey**
A PLACE	User-owned space · KV + SQL	**TinyCloud spaces**
A CONTRACT	Scoped, time-bound, revocable	**Delegations**
DISCOVERY	App and agent registry	**Registry**

We did not invent the parts. We assembled them in a shape that lets apps interoperate without giving up user ownership.

THANK YOU

If you are building apps where users should own their data, come talk to me.

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